ECE4824: *Digital VLSI-2 – IC Validation and Characterization*

1: VLSI Throwback - Crosstalk

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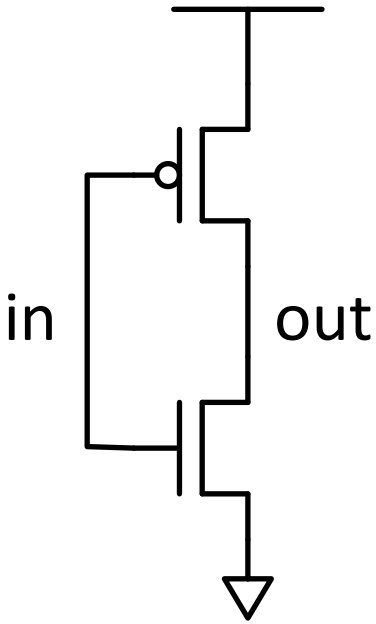
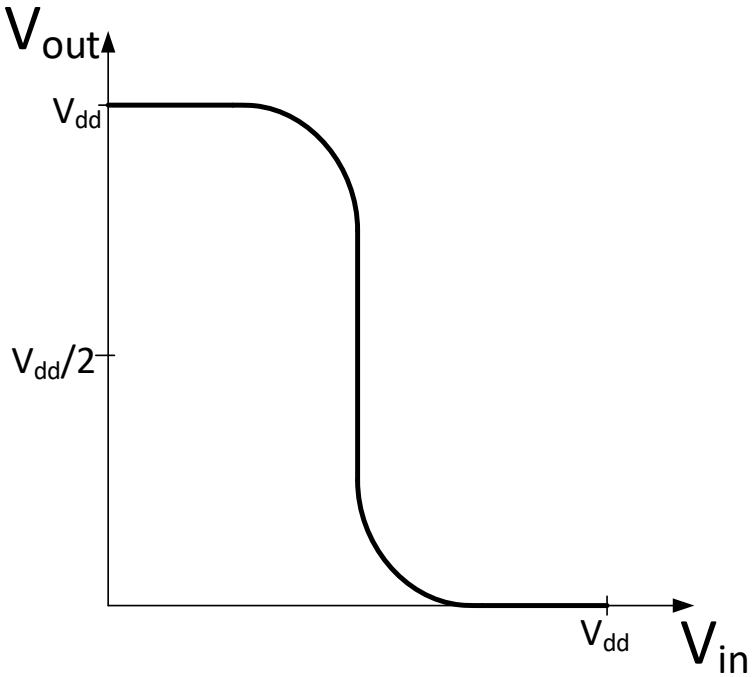
Georgia Tech

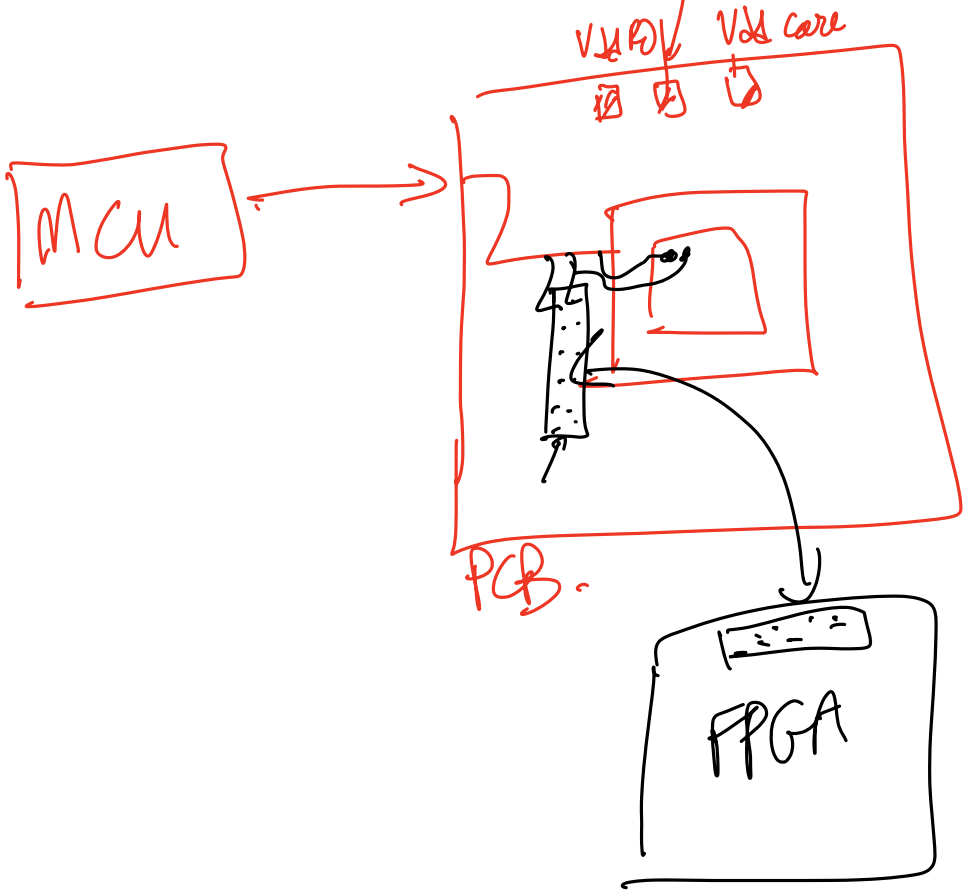
DC Transfer Function



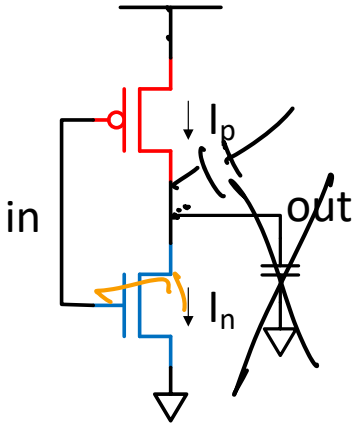


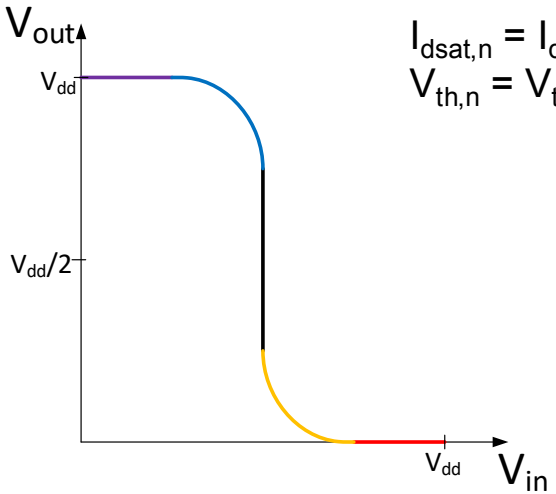
- DC transfer function of a balanced, ideal inverter
  - At each dc value of V_{in}, evaluate V_{out} and record

VDD/2

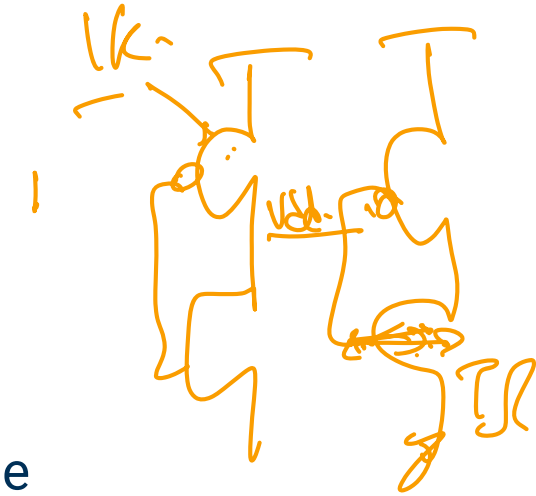

MCU
PCB.
FPGA

Analyzing the DC Transfer Function



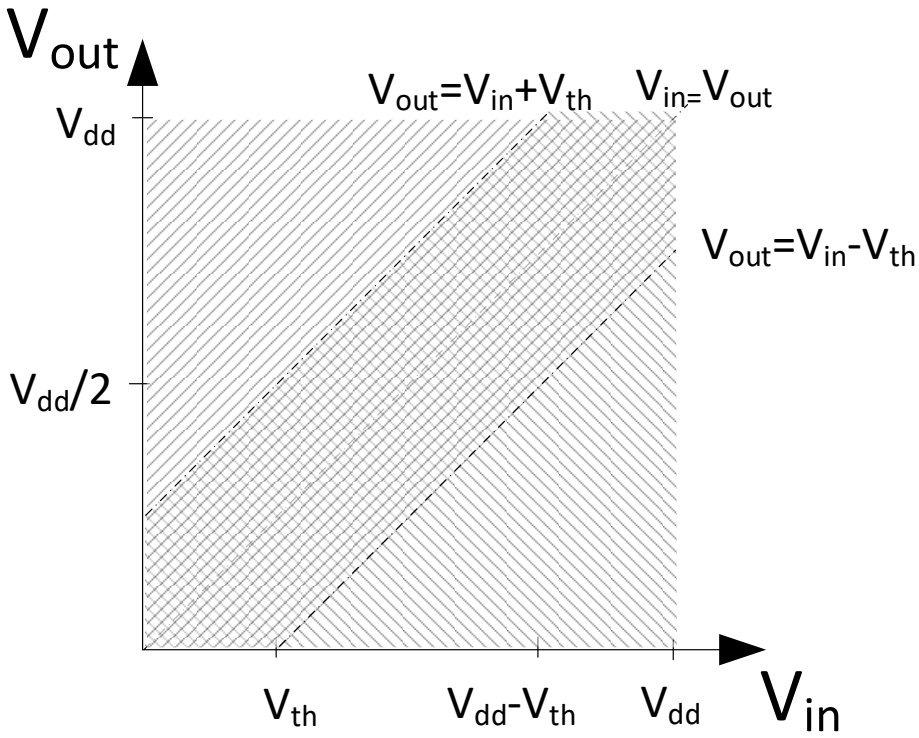


$I_{dsat,n} = I_{dsat,p}$
$V_{th,n} = V_{th,p} = 200mV$



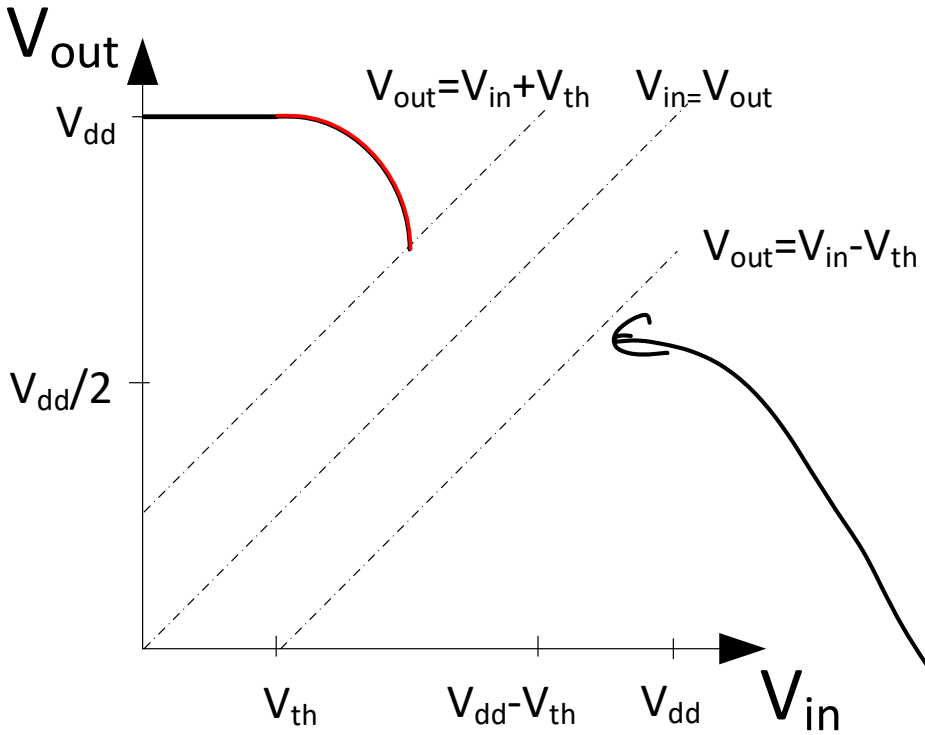
- Static analysis
  - Assumes load is capacitive (No steady-state current draw)
  - Voltages are dc (**Amount of capacitive load is irrelevant**)
- Inverter DC analysis
  - Devices operate over a variety of modes across voltage range
  - Governing equation: Pullup current == Pulldown current
  - Both, equation-based and graphical analysis is valuable
  - Useful in determining robustness to noise (Static Noise Margin/Holding Res)

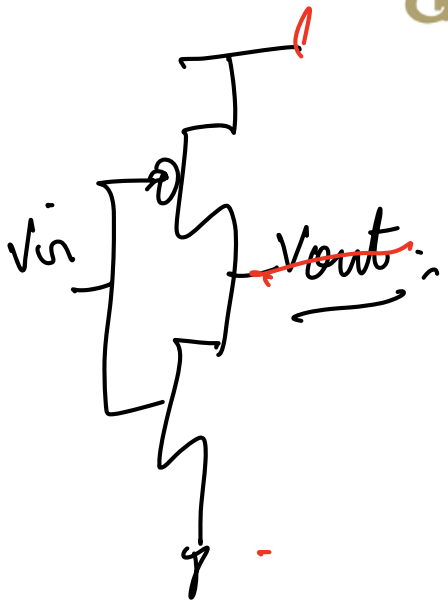
Inverter DC Analysis – Operation Zones



- $I_{dsat,n} = I_{dsat,p}$
- $V_{th,n} = V_{th,p} = 200mV$
- Mark out locus of operation

Inverter DC Transfer Function



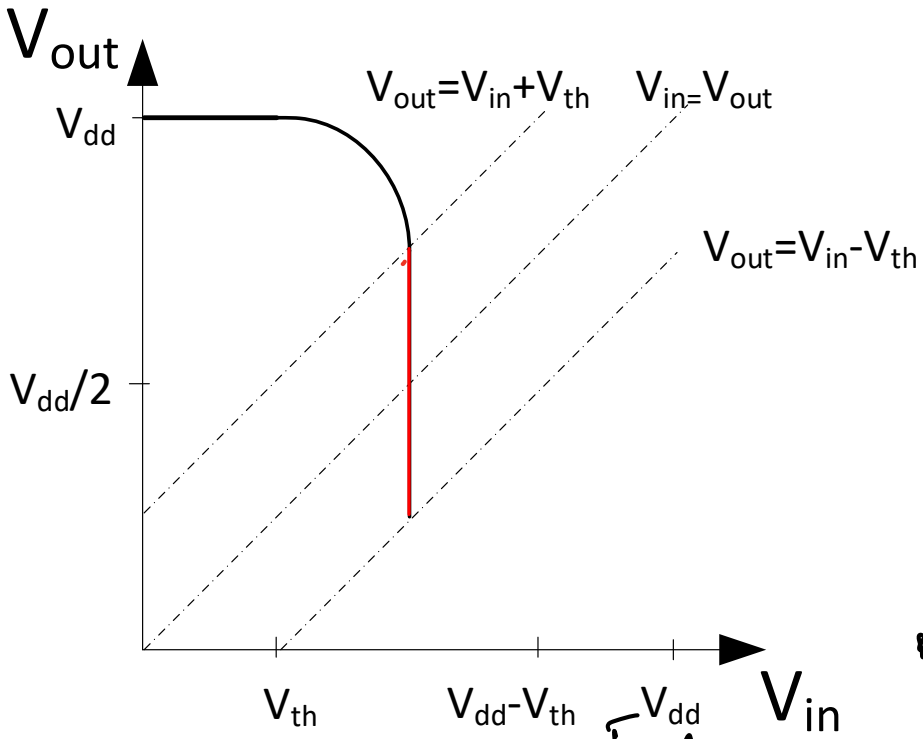


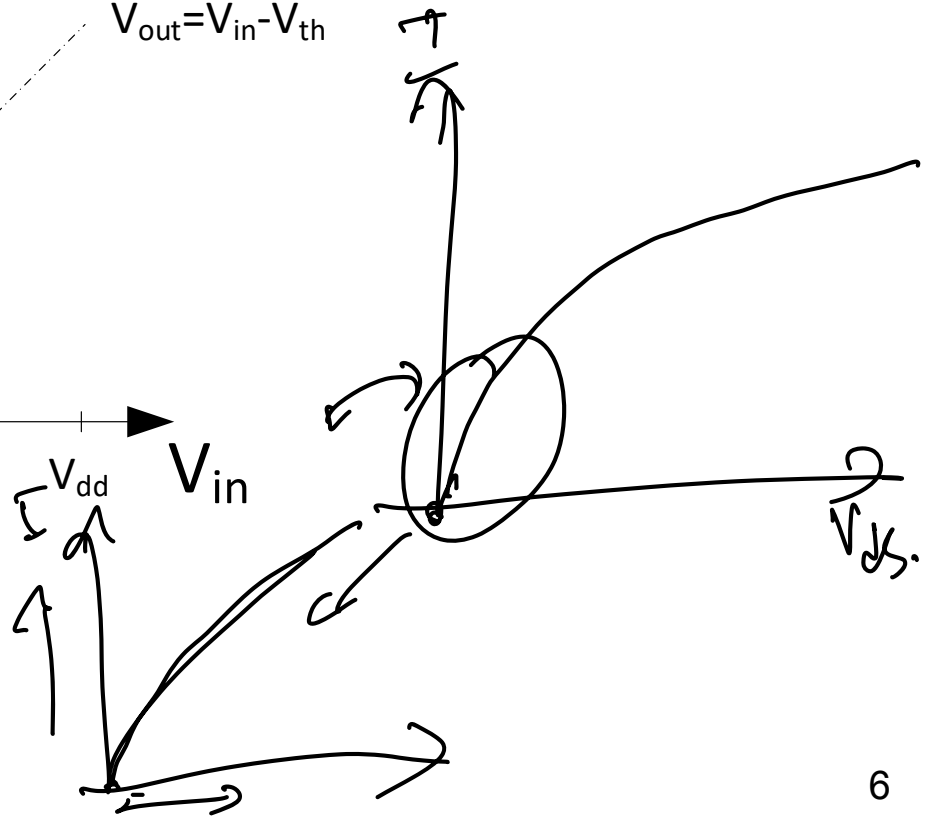
$V_{in} > V_{out} + V_{th}$

$V_{out} \leq V_{in} - V_{th}$

- $V_{th} < V_{in} < V_{out} - V_{th}$
  - Nmos turns on in saturation
  - Pmos in linear

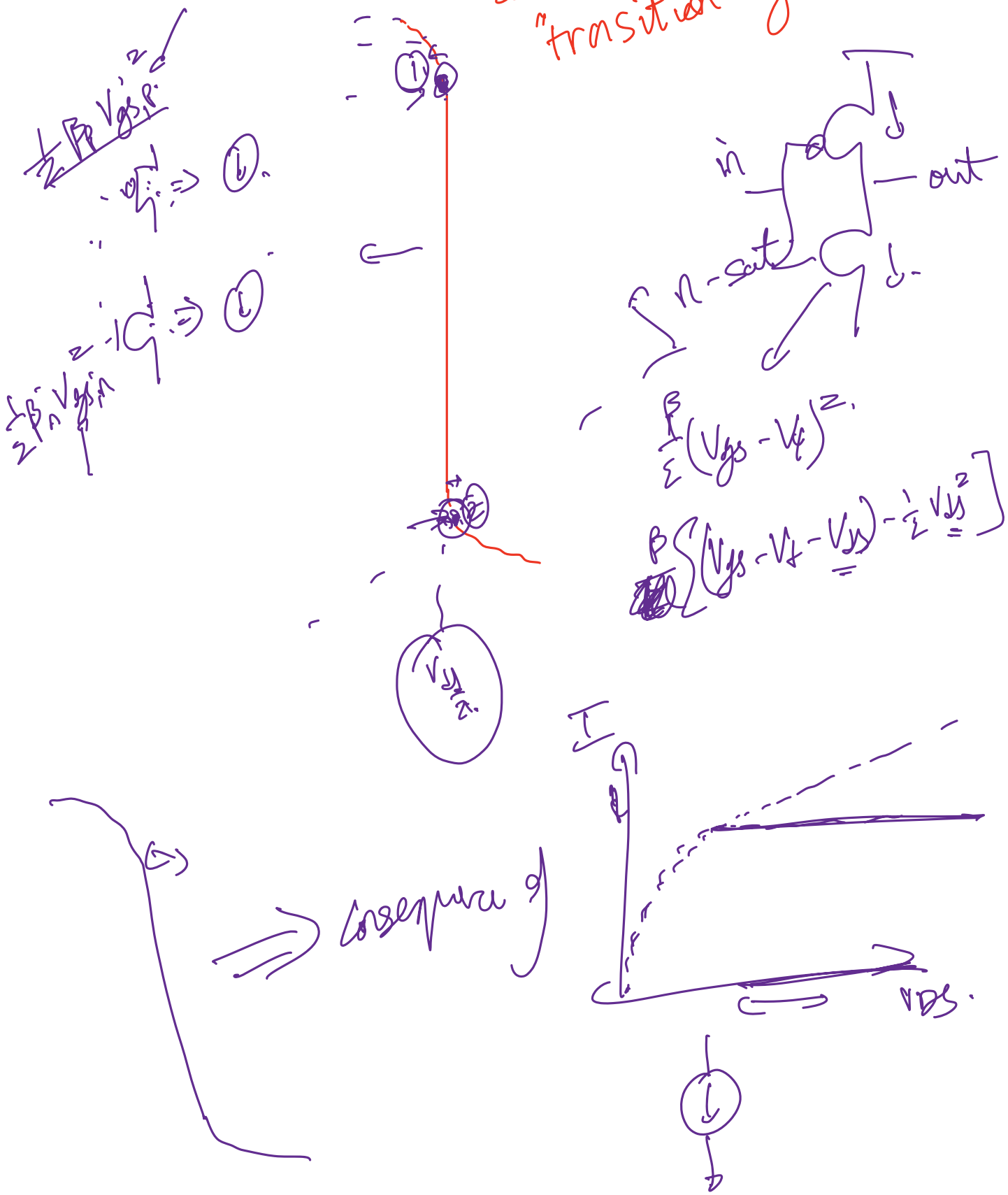
Inverter DC Transfer Function



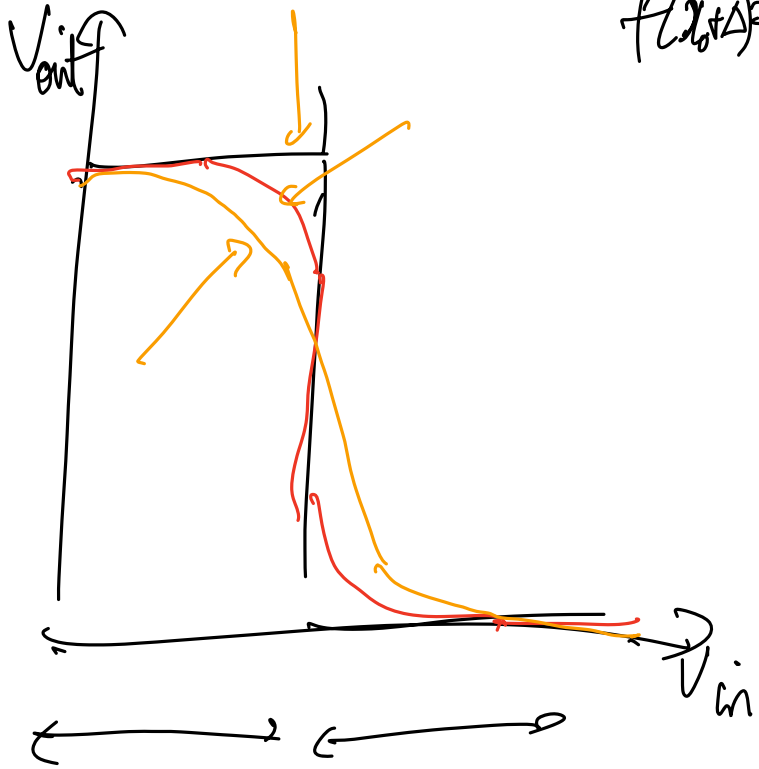


- $V_{out} - V_{th} < V_{in} < V_{out} + V_{th}$
  - Nmos remains in saturation
  - Pmos transitions to saturation

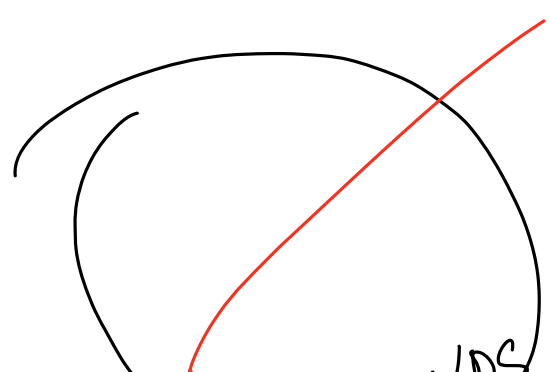
Zoom Into "transition region"



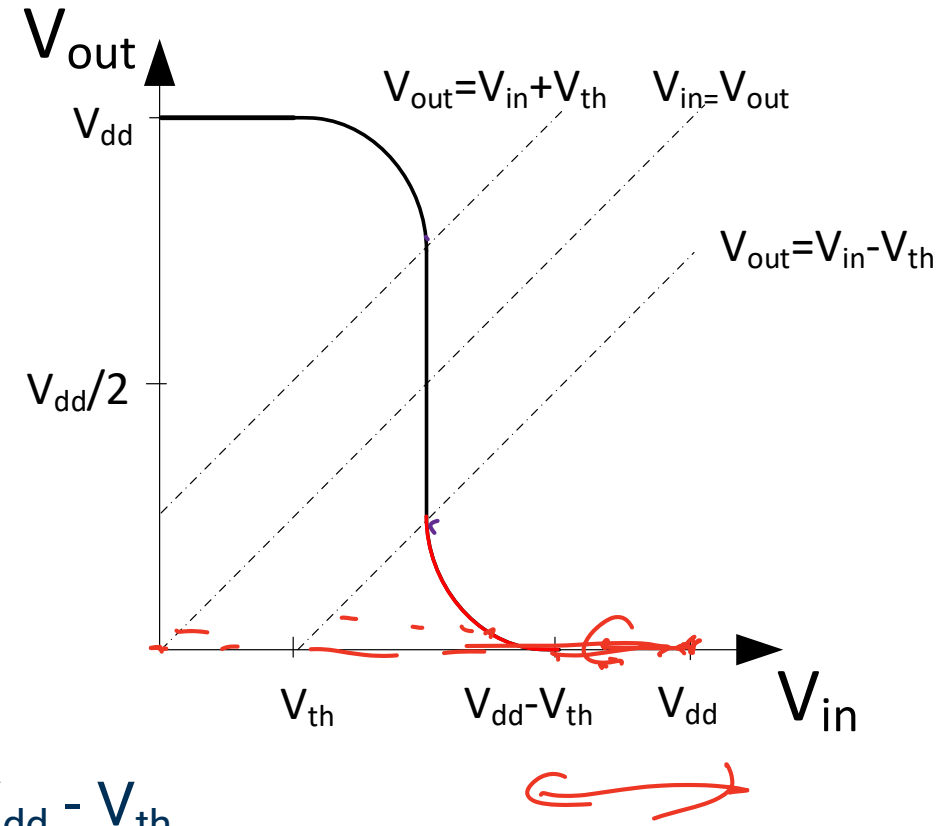
Consequence of

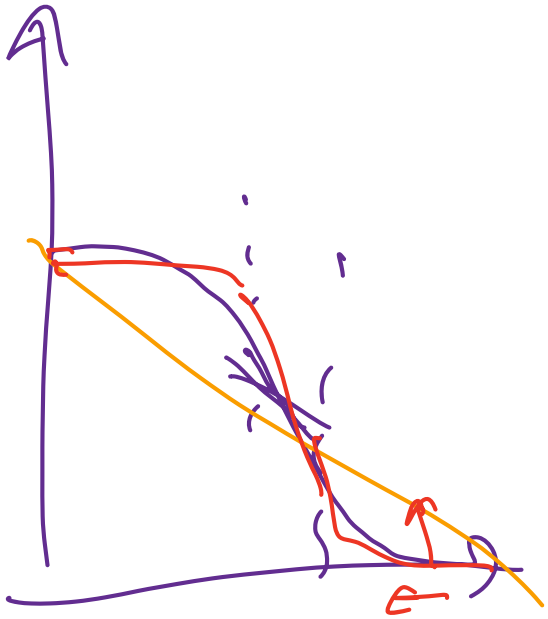


$$f(x_0+\Delta) = f(x_0) + f'(x_0)\Delta + \frac{1}{2}f''(x_0)\Delta^2 + \cdots$$



Inverter DC Transfer Function





- $V_{out} + V_{th} < V_{in} < V_{dd} - V_{th}$
  - Nmos transitions to linear
  - Pmos remains in saturation

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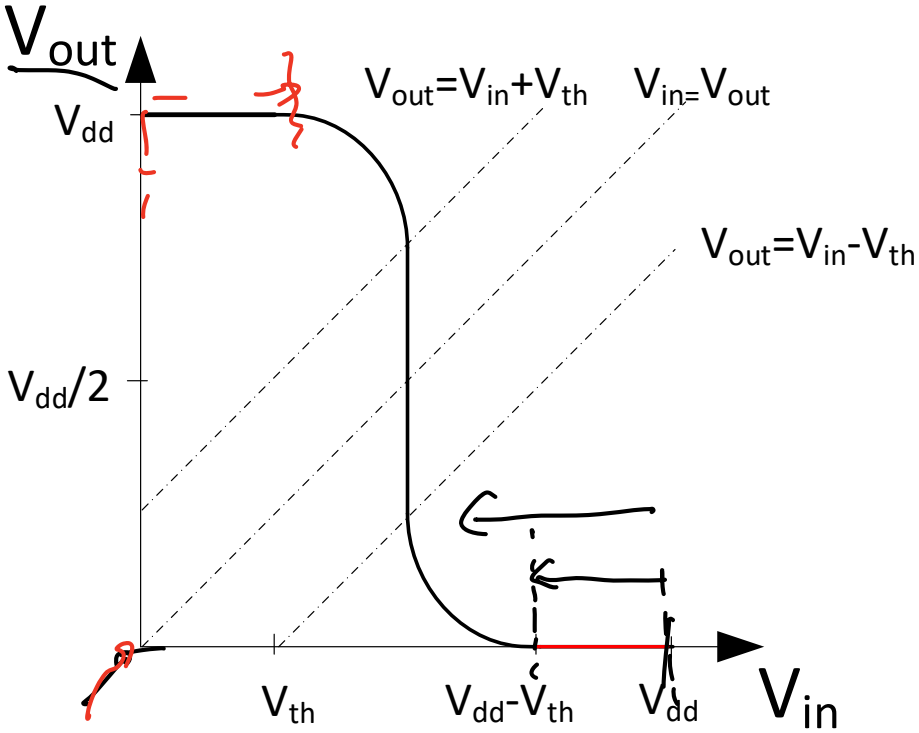
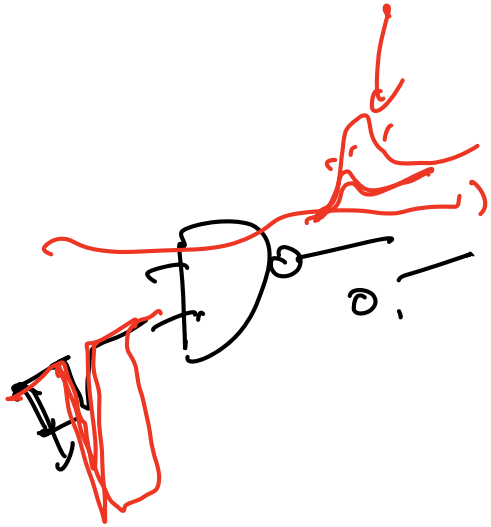
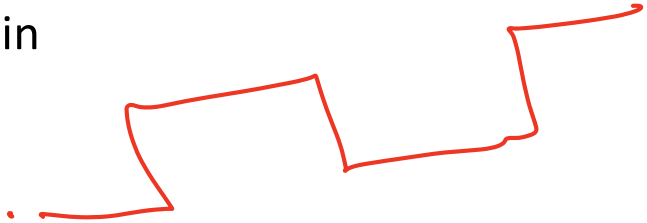
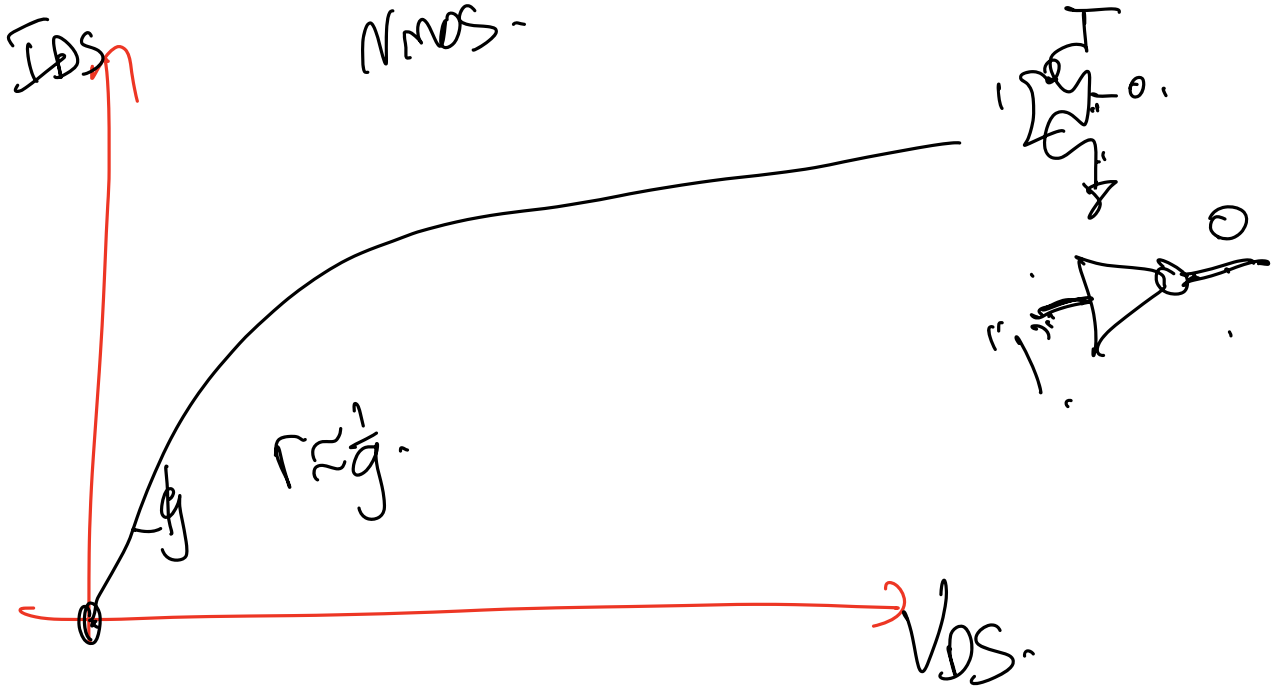
Inverter DC Transfer Function





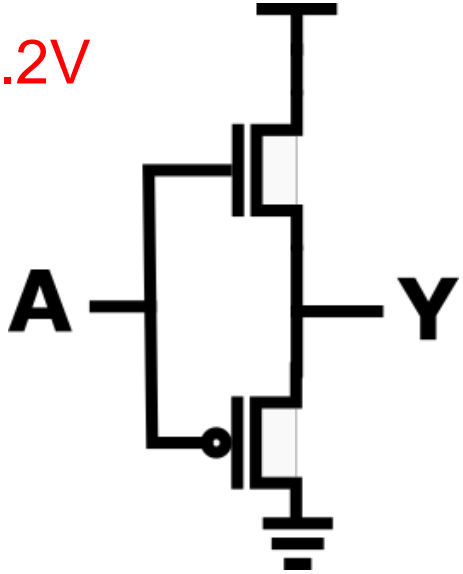


- $V_{dd} - V_{th} < V_{in} < V_{dd}$
  - Nmos remains in linear
  - Pmos transitions to cut-of

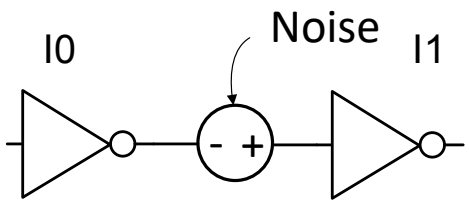

IDS
NMOS
g
r ≈ 1/g
VDS

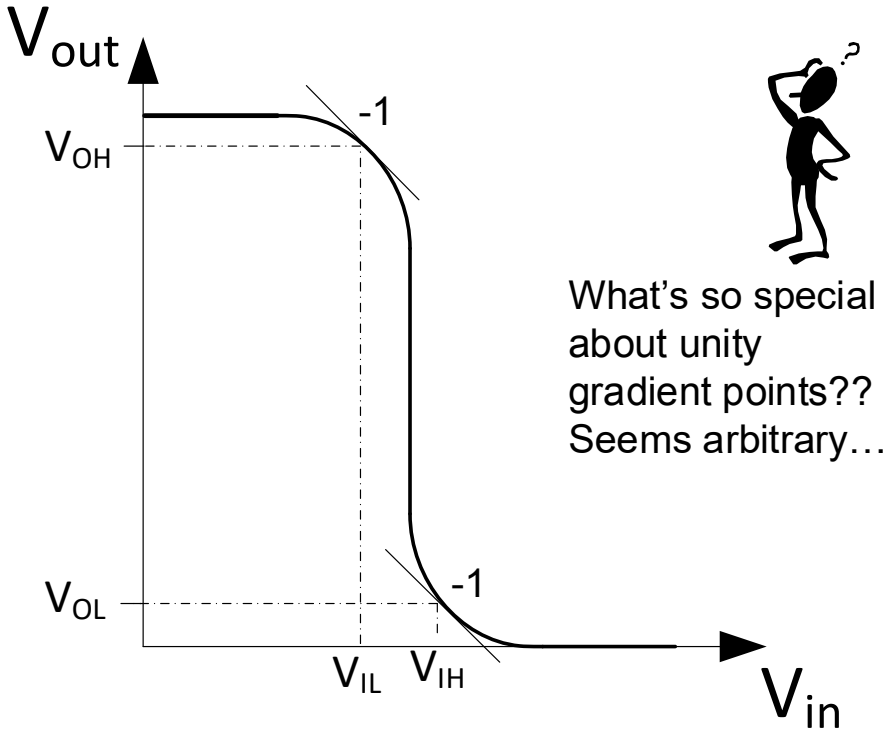
Inverter DC Transfer Function: Thought Exercises

- If $I_{dsat,p}=2*I_{dsat,n}$, sketch the the DC transfer curve relative to the balanced case
- Sketch DC curve for case where $V_{dd}=0.6V$, $V_{th} = 0.4V$
- Sketch DC curve for $V_{dd} = 1V$, $V_{th,n} = 0.4V$, $V_{th,p}= -0.2V$
- Sketch DC curve for circuit to the right (Trick question)



DC Transfer Function

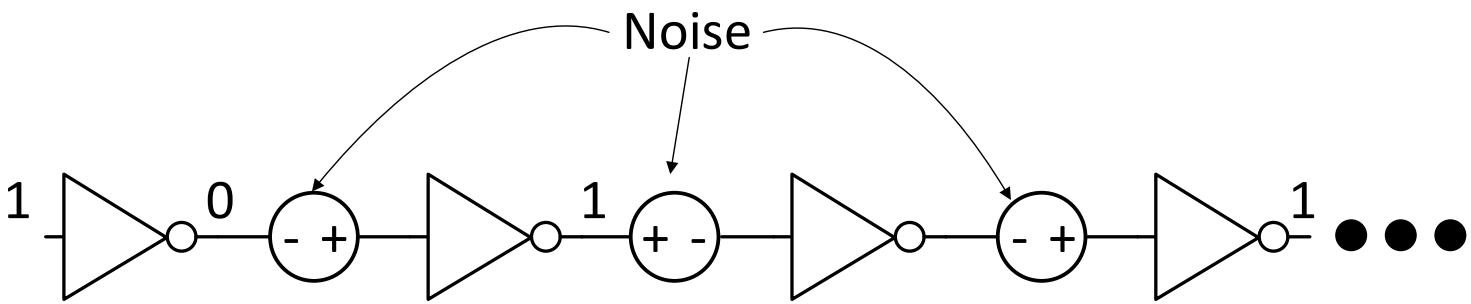




What's so special about unity gradient points?? Seems arbitrary…

- Static Noise Margin (NM)
- $NM_0 = V_{il} - V_{ol}$
- $NM_1 = V_{oh} - V_{ih}$
- What is the best possible NM for an inverter? What would it take?

DC Transfer Function



- Importance of unity-gain points (Cascade of inverters)
- What happens if noise voltage exceeds noise margin?
- Noise margin vs. V_{dd}

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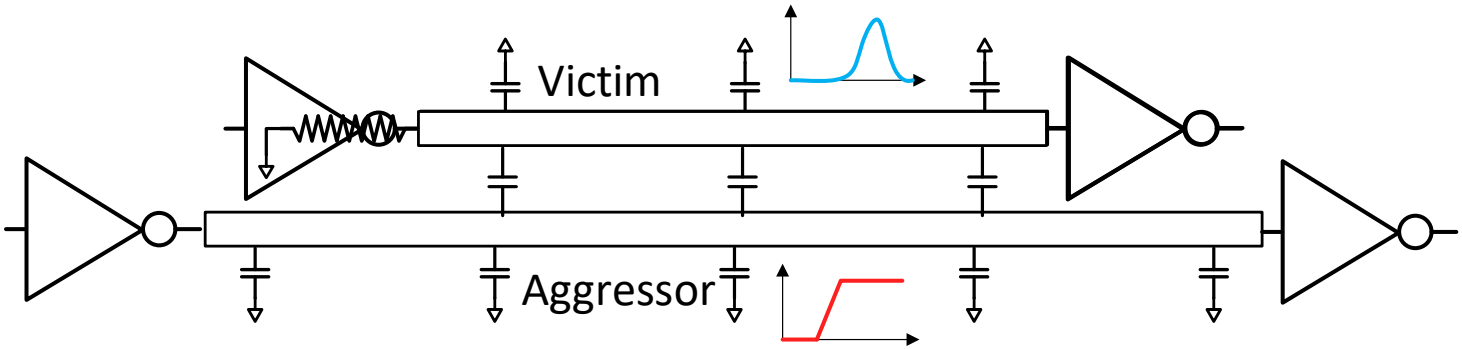
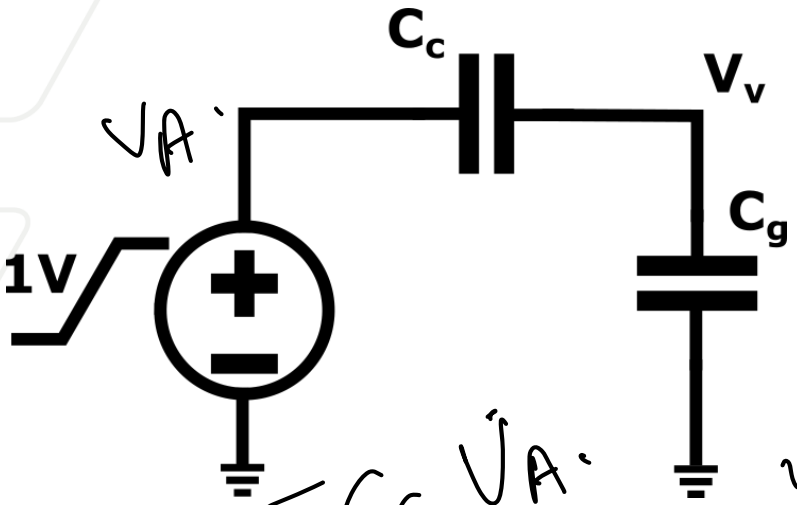
Crosstalk



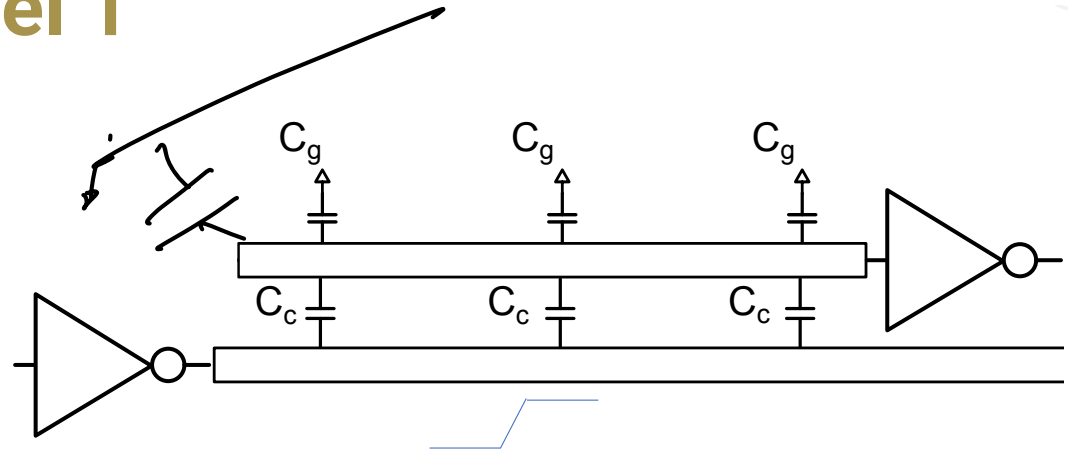
Why does the glitch matter?
What are the key glitch parameters
What are the relevant glitch-determining parameters

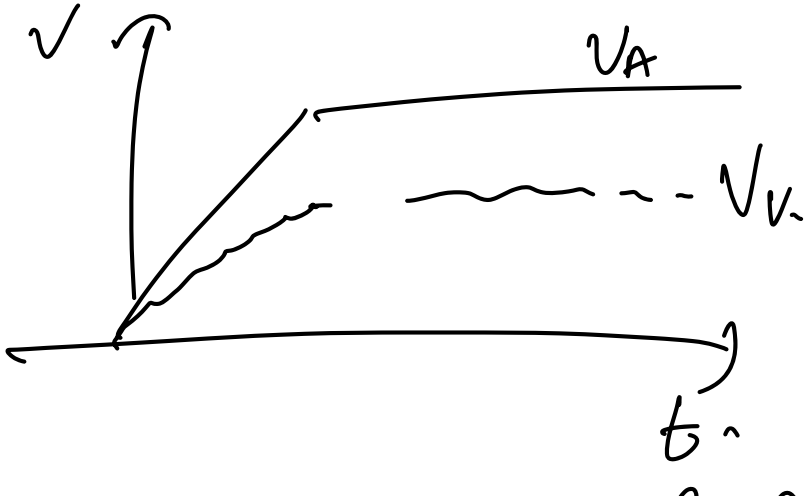
- Static inverter analysis also important for holding resistance analysis
  - Crosstalk analysis
  - Holding resistance impacts peak noise glitch

Crosstalk (Simpler) Model 1

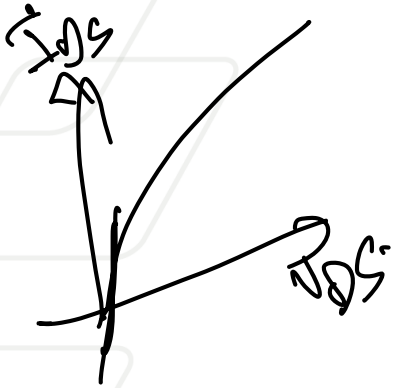


$$V_v = \frac{C_c V_A}{C_c + C_g}$$



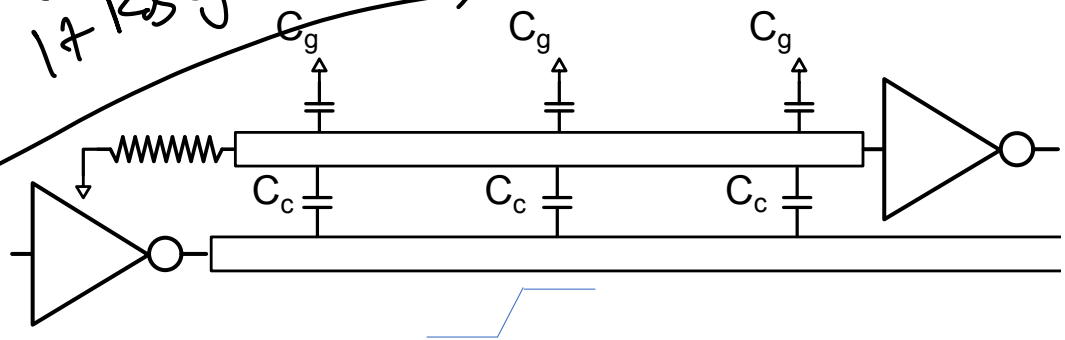


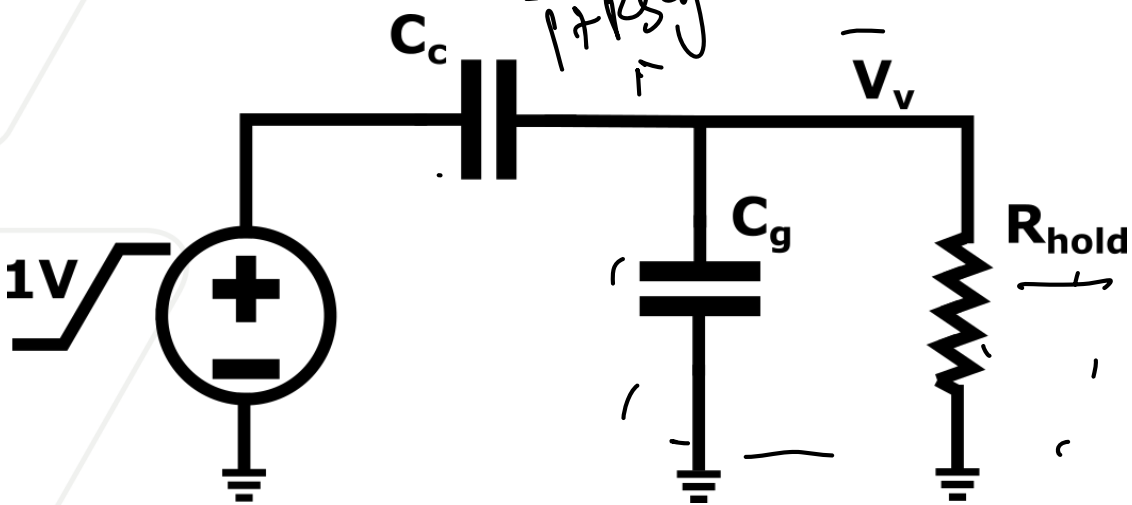
Crosstalk Model 2

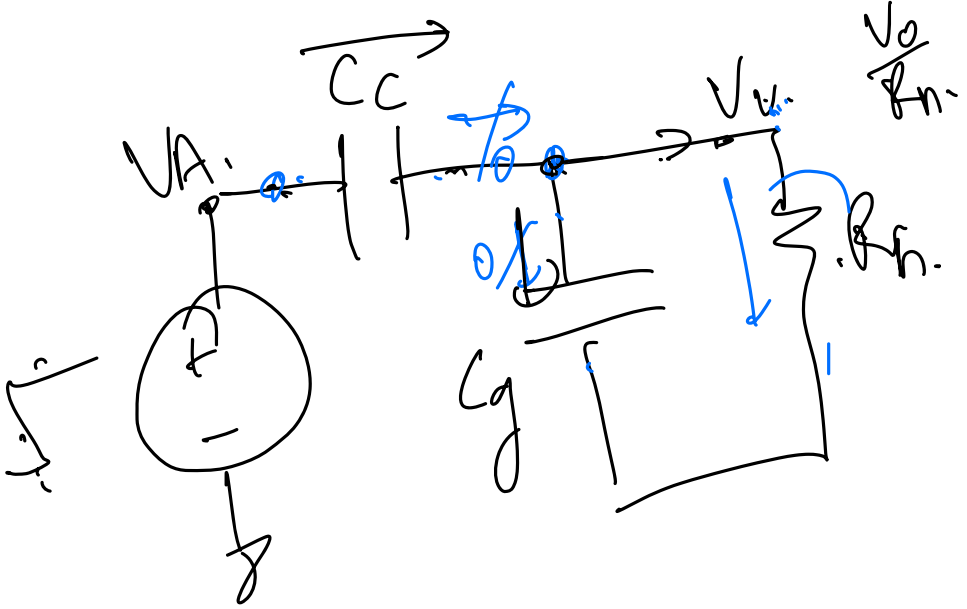


$$\frac{R}{1+R_sC_g} \Rightarrow \frac{R_sC_c}{1+R_sC_g+R_sC_c} = \frac{R_sC_c}{1+R_s(C_c+C_g)}$$

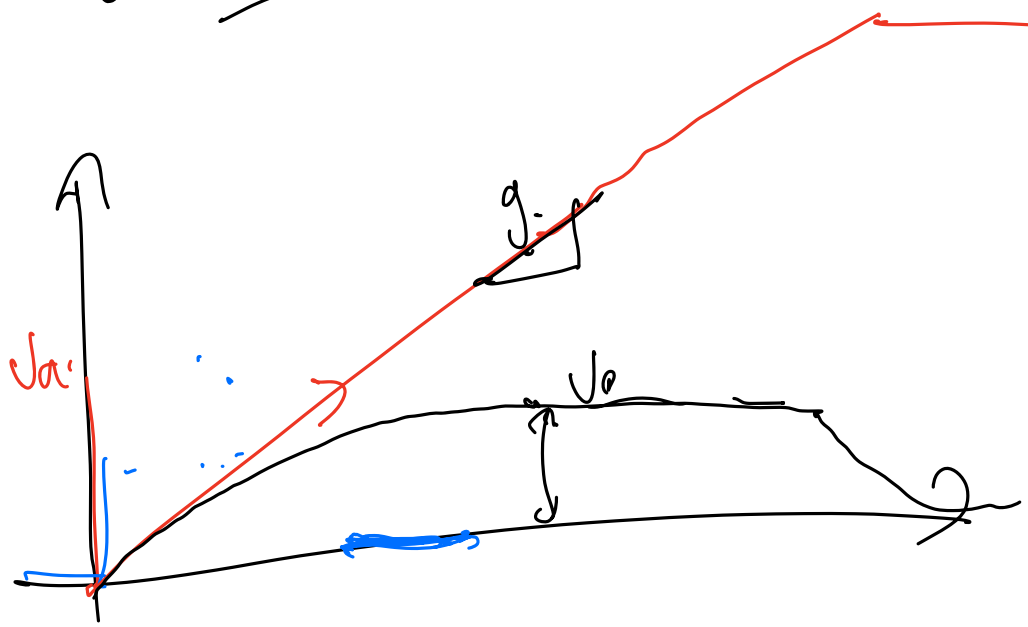
$$\frac{R}{1+R_sC_g} + \frac{1}{sC_c}$$

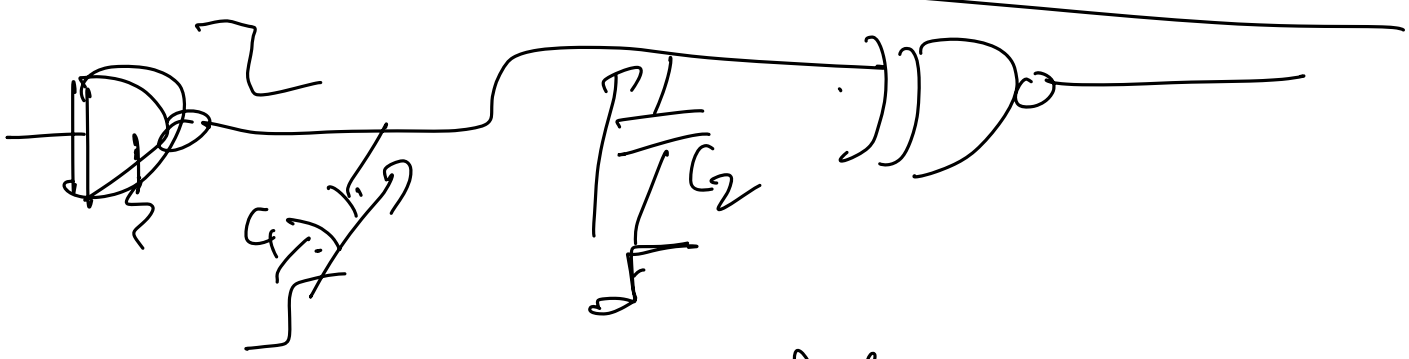






$I_{cc} = C \frac{dV}{dt}$





Usual:

$C_p = C_{tot}$

$\Delta V = V_{dd}$

$Q = C_{tot} V_{dd}$

$= C_{tot} V_{dd}$

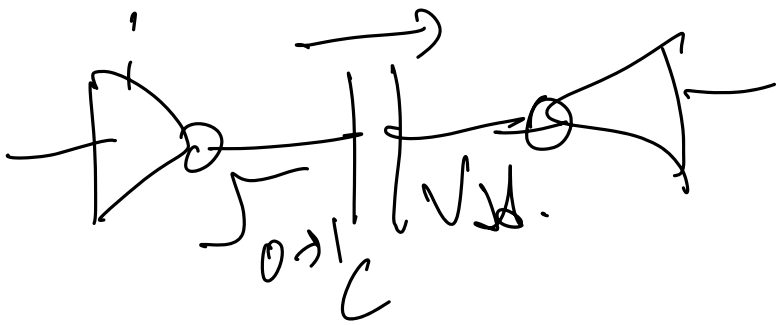
Agg.

$C_p = C_{tot}$

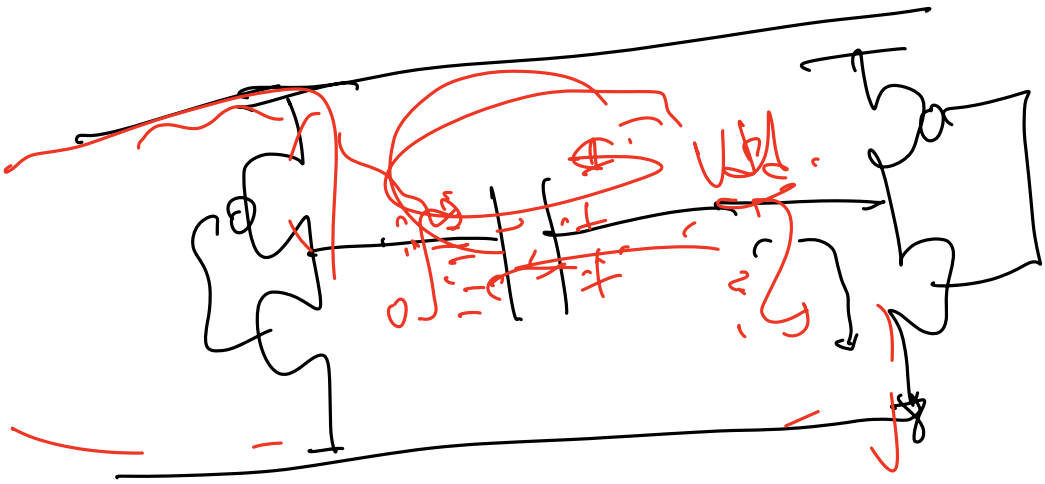
$\Delta V = V_{dd}$

$Q = C_{tot} V_{dd} + \sum C_i V_{dd}$

$\tau = \frac{Q}{I_{drive}}$



CVDD.

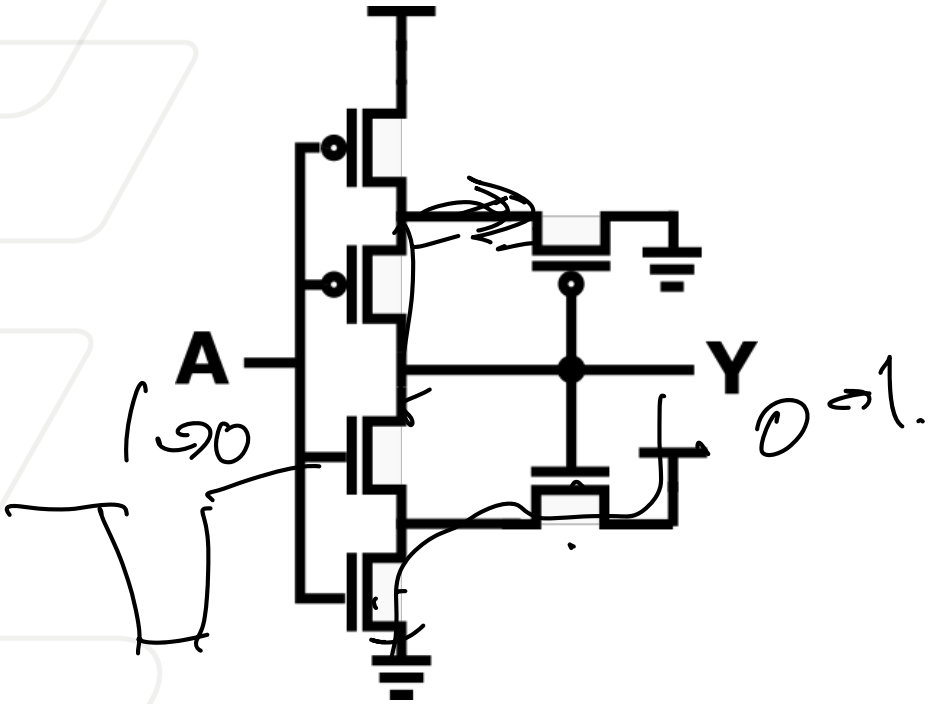


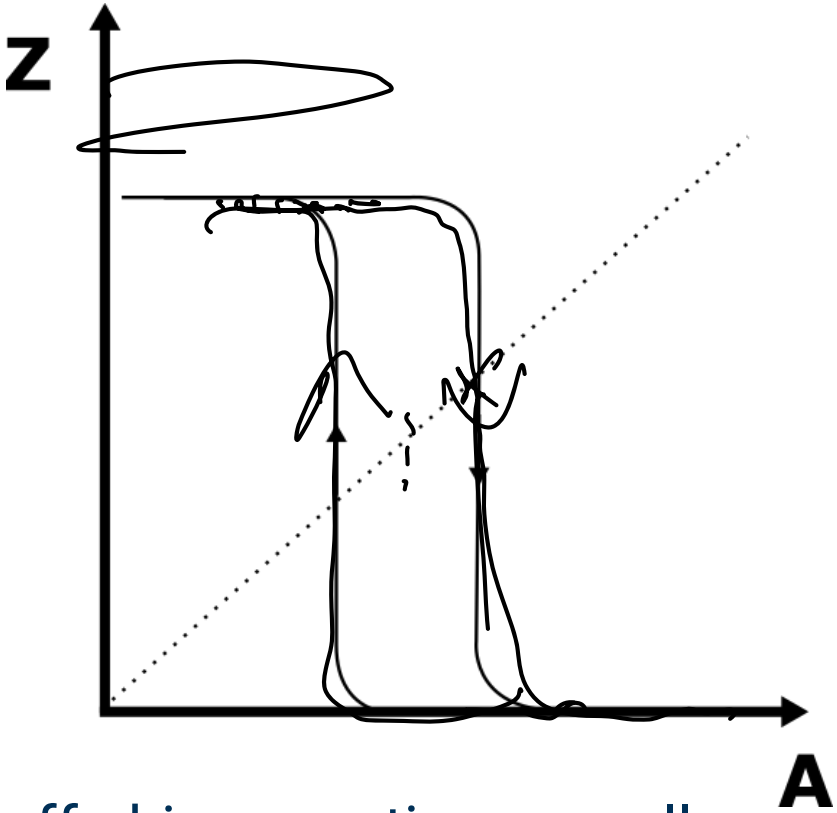
PVT effects of X-talk

- Model 1:
  - V_a has a higher voltage swing…
  - But victim has a higher noise margin
- Model 2:
  - $\downarrow V_{dd} \rightarrow \downarrow I_d \rightarrow \downarrow \frac{dV_a}{dt}$ (lower aggressor slew) AND
  - Wire-resistance degradation less important….. BUT

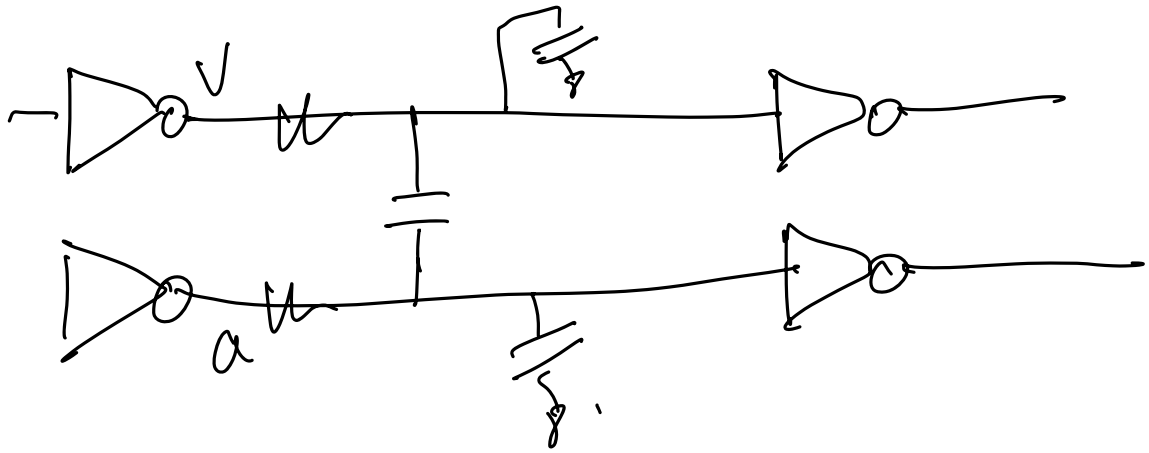
  - But, lower noise margins
  - Higher holding resistance
- Which dominates?
  - Try it out!
  - Clue: Evaluate across different V_{dd} , and for varying driver strengths and wire resistances…

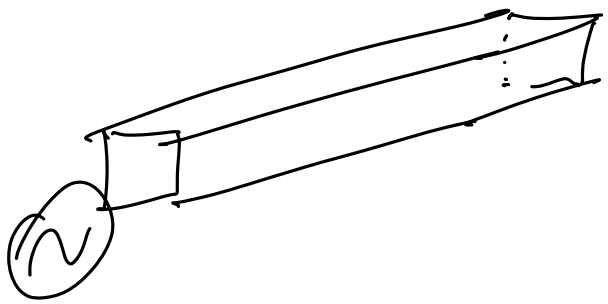
Crosstalk considerations





- Not restricted to on-chip – impacts off-chip operation as well
  - Long runs of closely spaced wires with no ground-plane/coupling around….
- Motivates use of Schmitt triggered inputs (hysteresis)
  - What does a Schmitt trigger DC Noise margin look like?





$R = 1k\Omega$

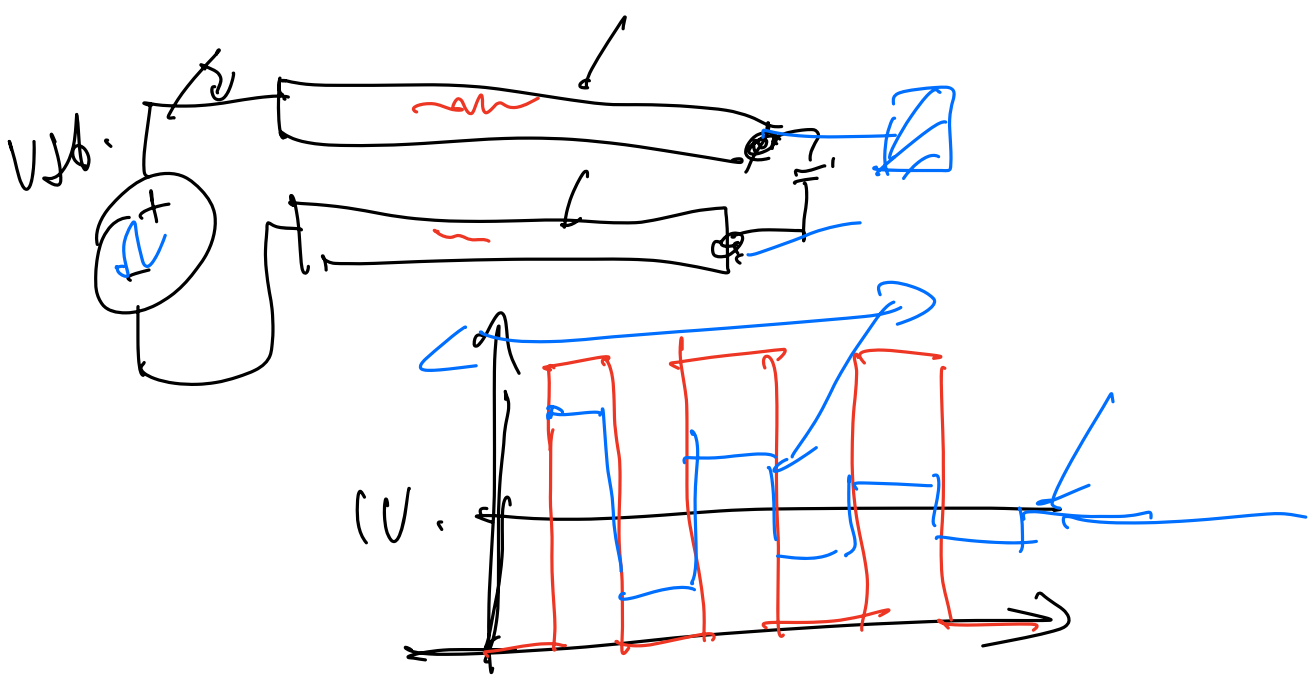
$L = 2A$

$f = 10G$

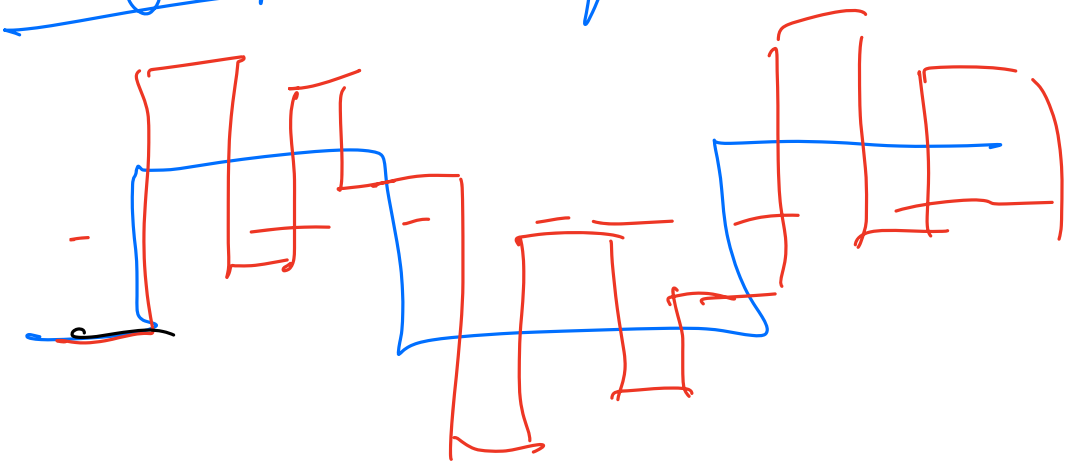
$|Z_L| = \omega L$

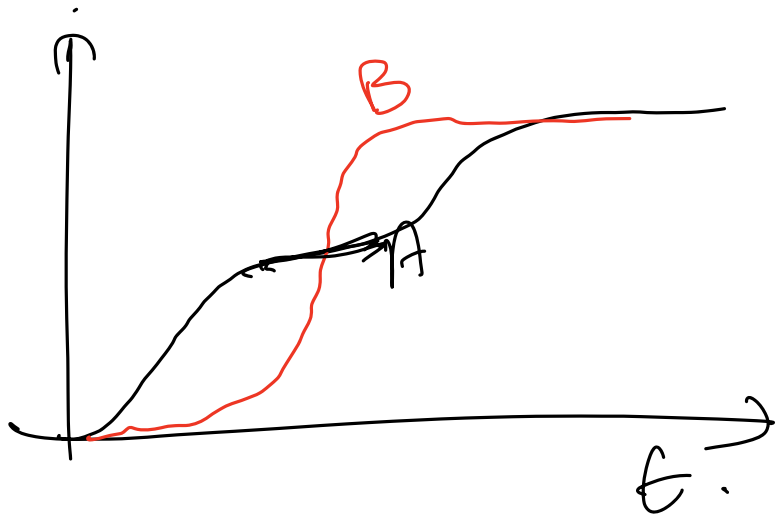
$= 62.8 \times 2$

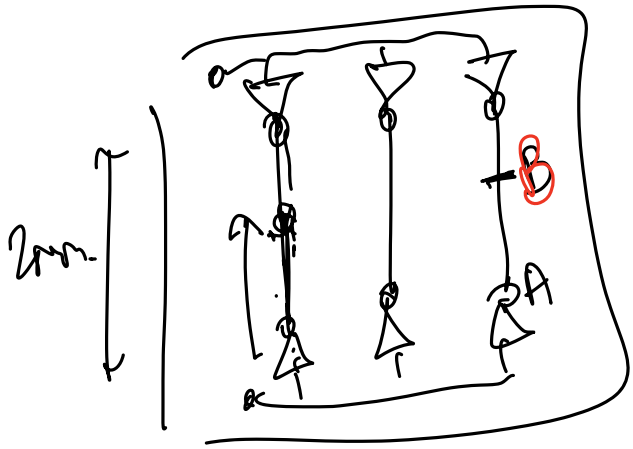
$= 125\Omega$

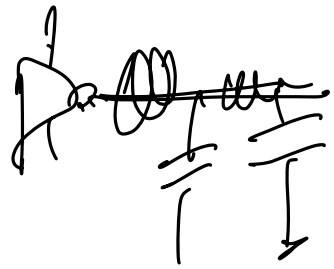


Telegraphers Equations.









4PS.